

DISCOUNT STRATEGY

FROM INSIGHT TO IMPACT:

"A data-driven approach to identifying at-risk customers and maximising discount ROI at a premium direct-to-consumer apparel retailer."



By : Paula Moreno



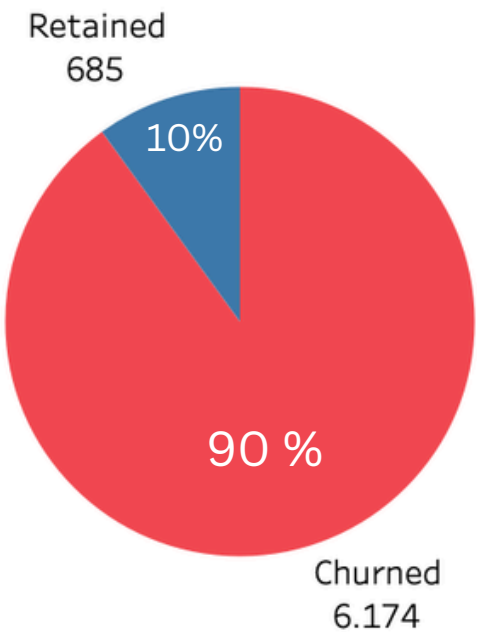
Brief summary

"The current discount strategy is not fixing retention and our highest-value customers are the least loyal."

The Retention Crisis

90% of customers never return. Each retained customer is worth 2.5x more (\$114.75 vs \$46.13). The current discount strategy is not fixing this.

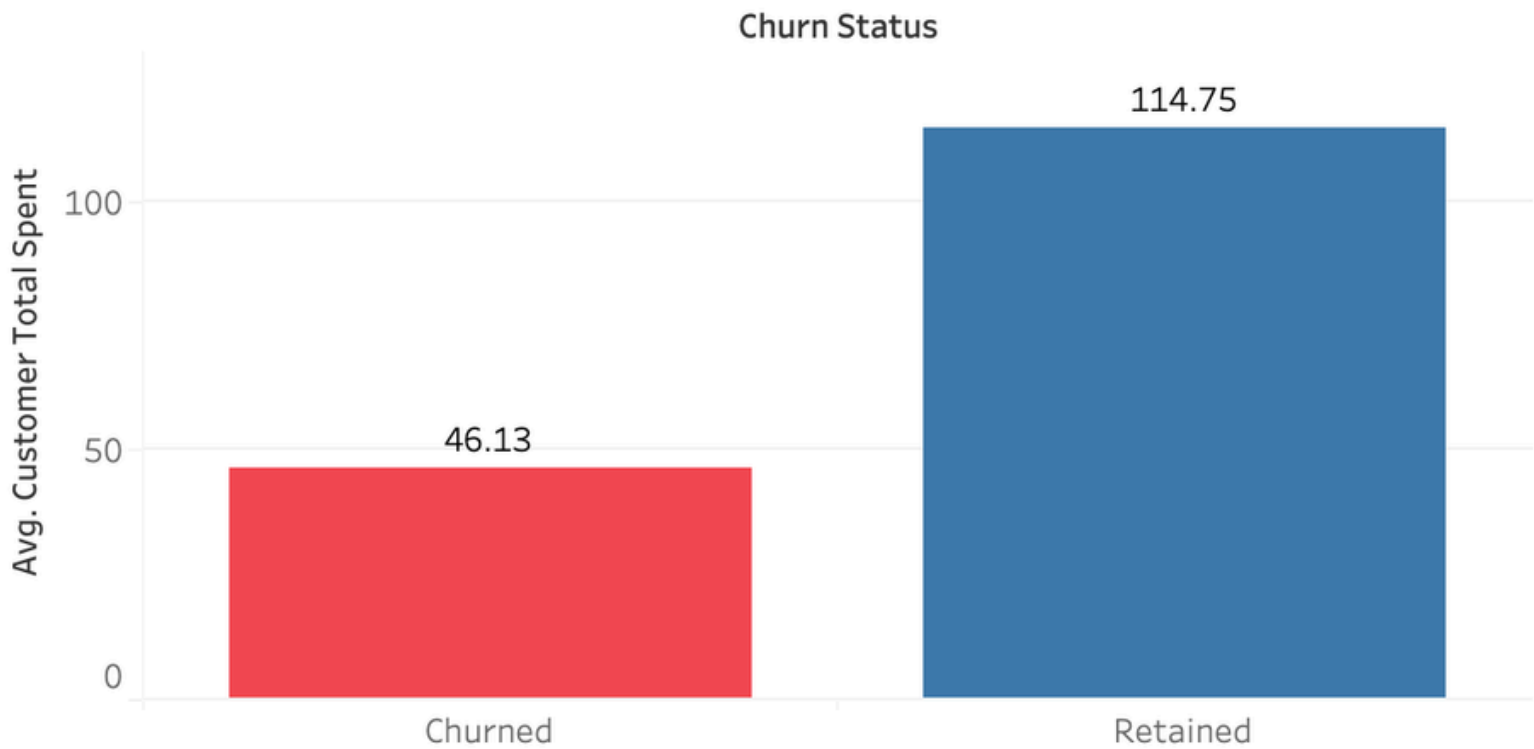
Churned vs Retained Customers



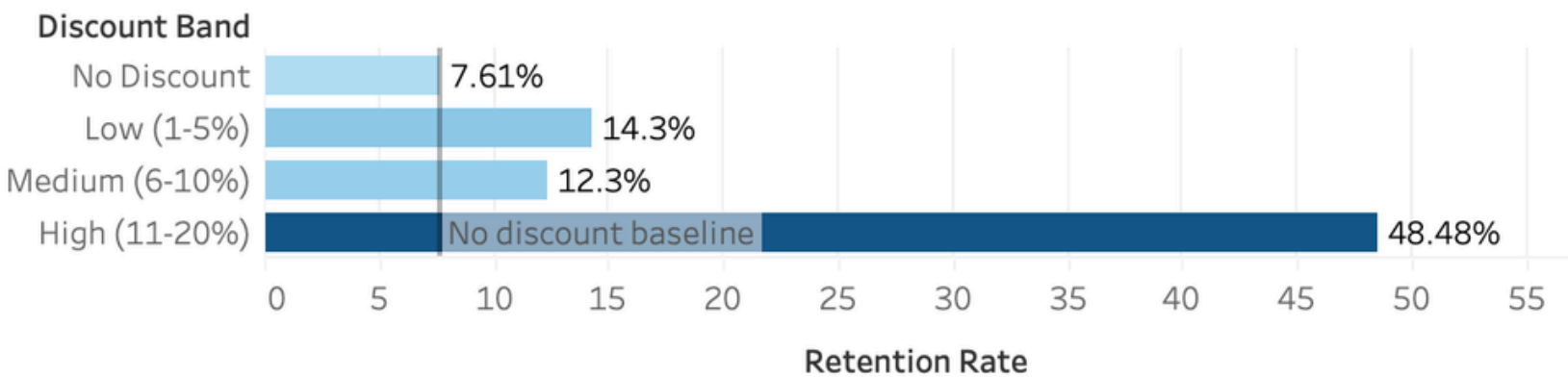
Churn Status

- Churned
- Retained

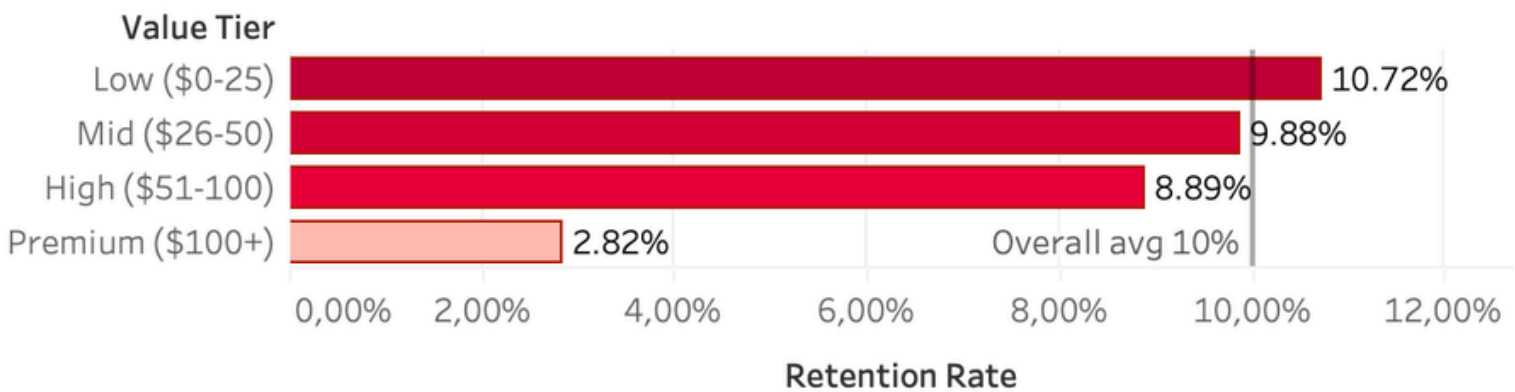
Avg Lifetime Spend per Customer — Churned vs Retained



Retention by Discount - Does Discount Depth Affect Retention?



Retention Rate by First Order Value Tier (%)



Exploratory analysis *and* deeper findings

Block 1: Understanding Customer Behaviour

Q1. What does a returning customer look like vs. a churned one?

Result:

Retained customers spend **2.5× more than churned**

Q2. Does discount depth affect whether someone returns?

Result:

Only high discounts lift retention **48.5% vs 7.6% baseline**

Q3. Which customer segments have the worst retention?

Result:

Premium spenders barely return **2.82% retention \$127 spend**

Block 2: Understanding Customer Value

Q4. How does customer value differ across discount behaviour groups?

Result:

Always discounted = lowest loyalty, **5.0% retention vs 8.0% never discounted**

Q5. What drives how much a customer spends over their lifetime? Correlation analysis.

Results:

First order value is the key driver, **Premium tier spends \$142 lifetime**

Block 3: Business Impact

Q6. What is the revenue at risk from churned customers?

Result:

Discount budget largely wasted **\$4,207 spent on churned (74%)**

10% retention lift = **\$58,870 additional revenue**

That is a **14× return on wasted budget**



Key Insights



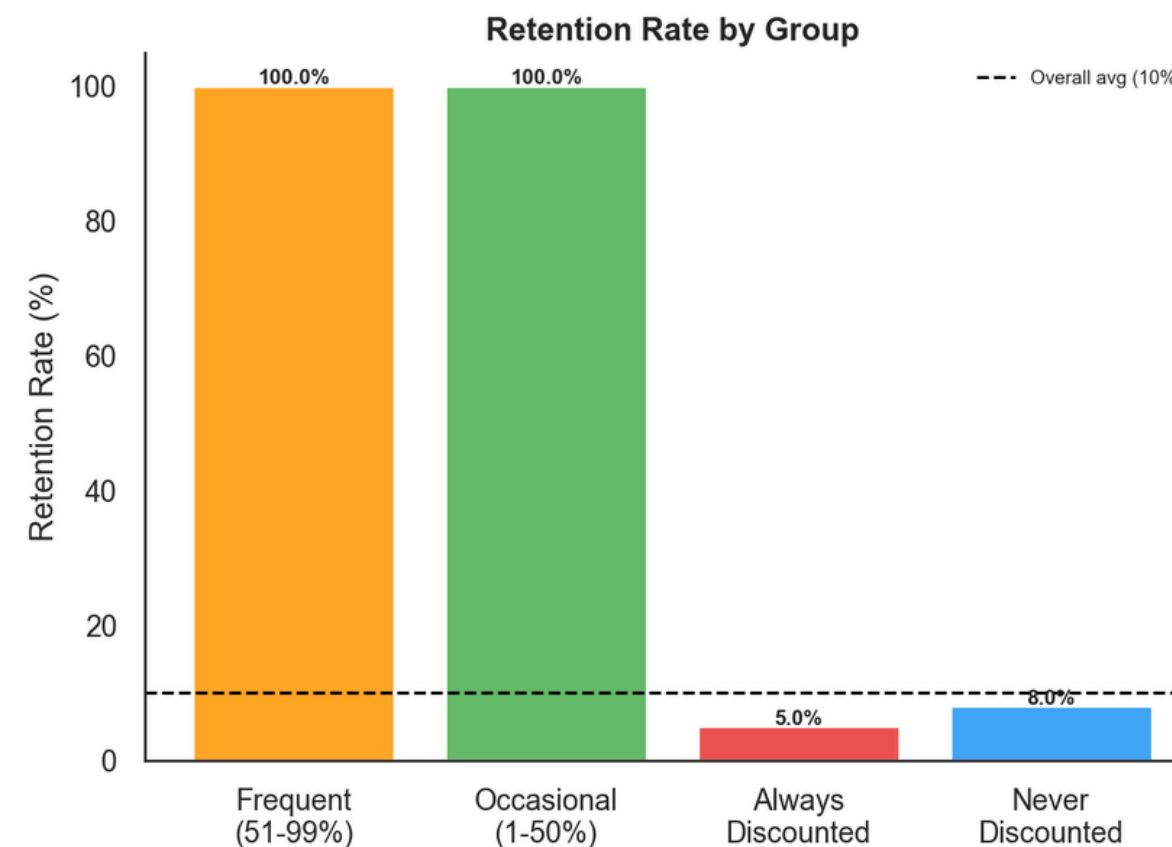
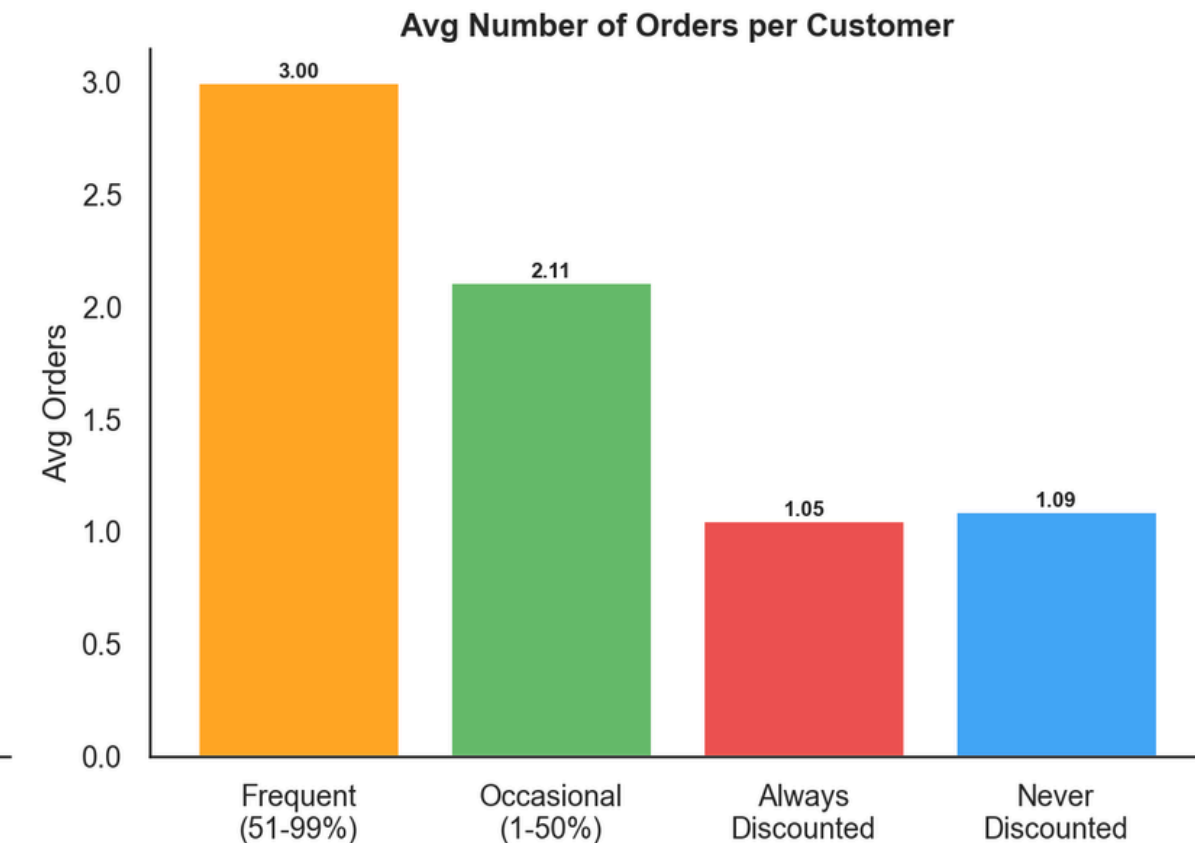
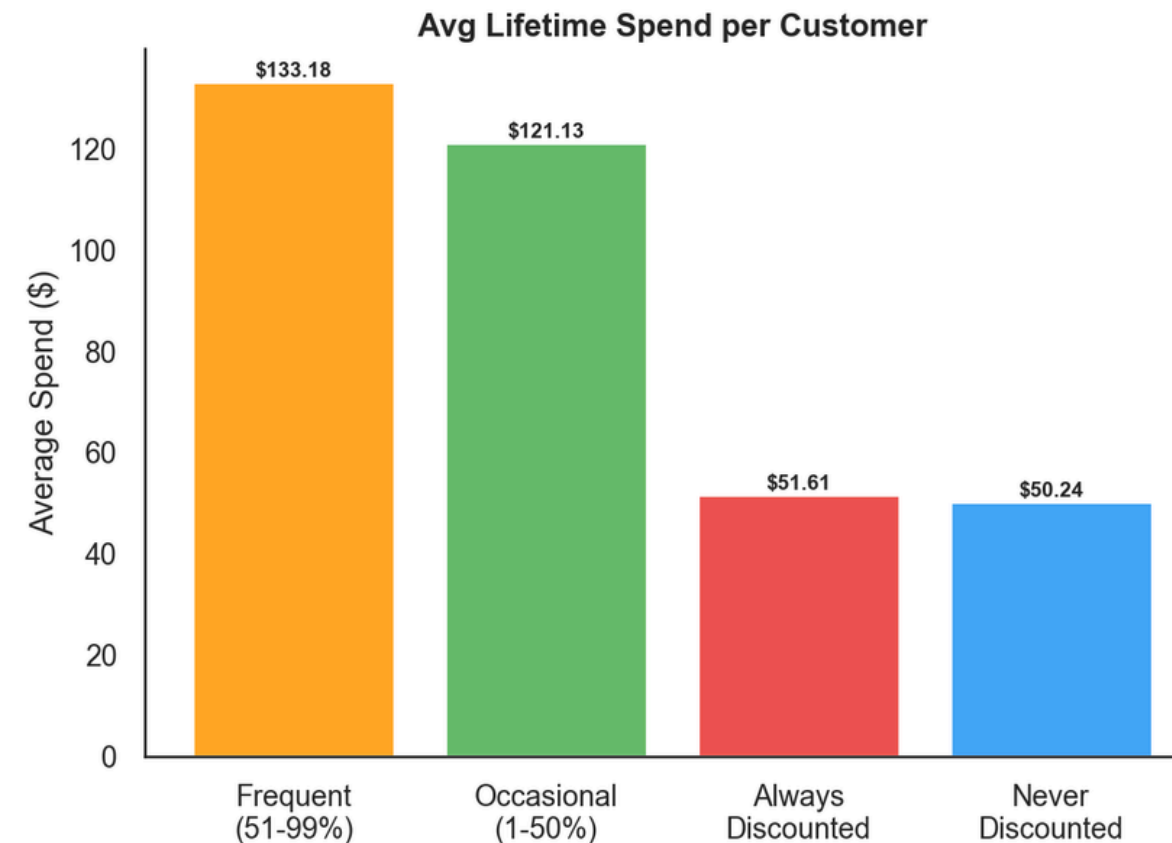
visual deep-dive into the key metrics related to our discount strategy

MOST POWERFUL FINDING FROM THE ENTIRE EXPLORATORY ANALYSIS

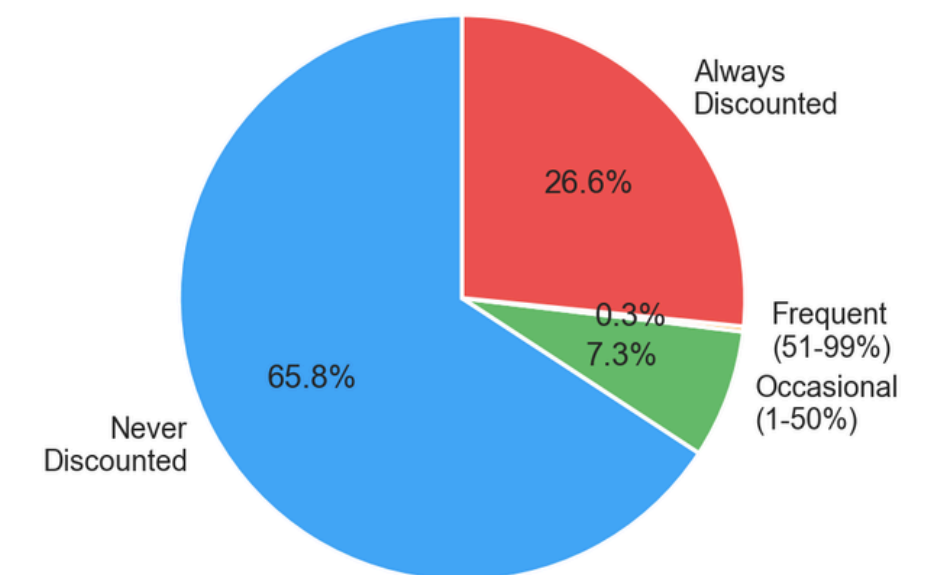
Most powerful Q4 insights:

1. Always Discounted customers have LOWER retention (5.0%) than Never Discounted customers (8.0%). The business is giving discounts to 1,873 customers who return less than customers who received no discount at all. This completely destroys the justification for the current discount strategy.
2. Occasional users (100% retention, \$121.13 spend) vs Always Discounted (5% retention, \$51.61 spend). That is a 20% difference in retention and 2.6x difference in spend between these two groups.
3. Never Discounted customers generate 65.8% of all revenue. The group that receives zero discounts generates nearly two thirds of the business's total revenue.

Q4: Customer Value Across Discount Behaviour Groups



Revenue Share by Discount Group



Key Insights

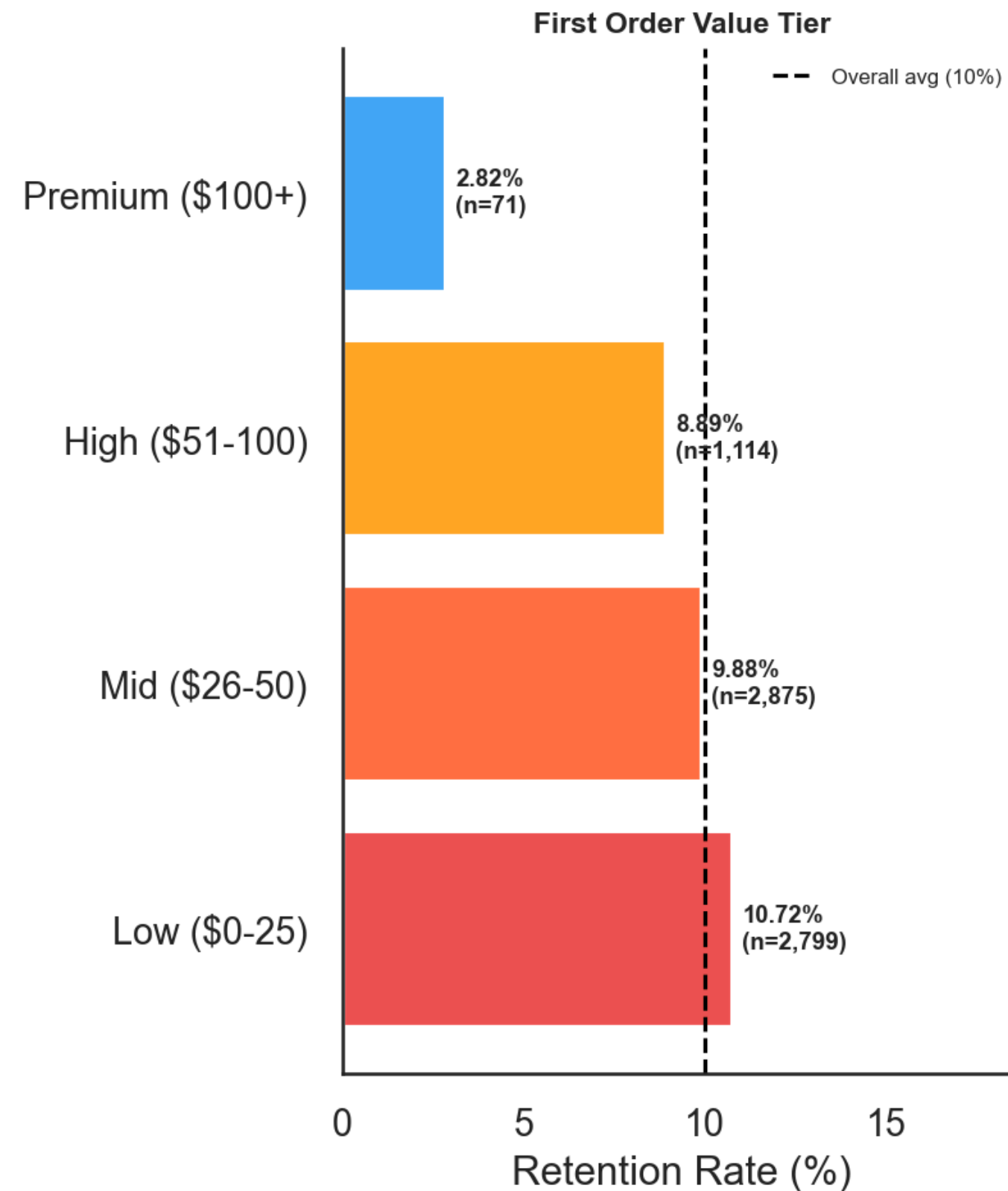


visual deep-dive into the key metrics related to our discount strategy

Most powerful Q3 insights:

1. Premium customers (71 people) return at only 2.82%. But look at their average spend, \$126.88 the highest of any group. These are genuinely high-value customers who almost never come back. This is your most shocking finding.
2. Mid tier is the largest group (2,875 customers) and generates 45.3% of all revenue. Yet their retention is below average at 9.88%. This is where the biggest revenue opportunity sits, not in the tiny premium group.
3. Low spenders actually have the BEST retention at 10.72%. This is counter-intuitive our cheapest customers are our most loyal.

Q3: Which Customer Segments Have the Worst Retention?

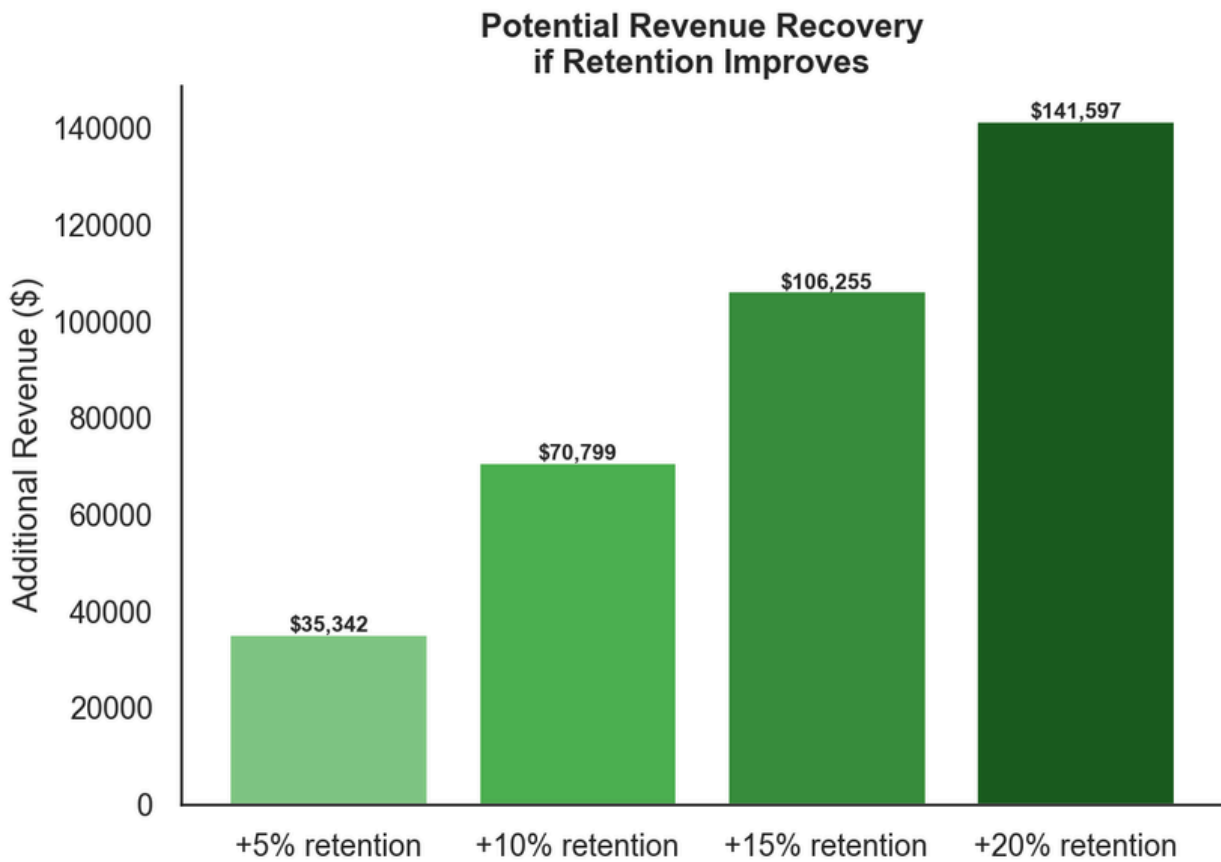
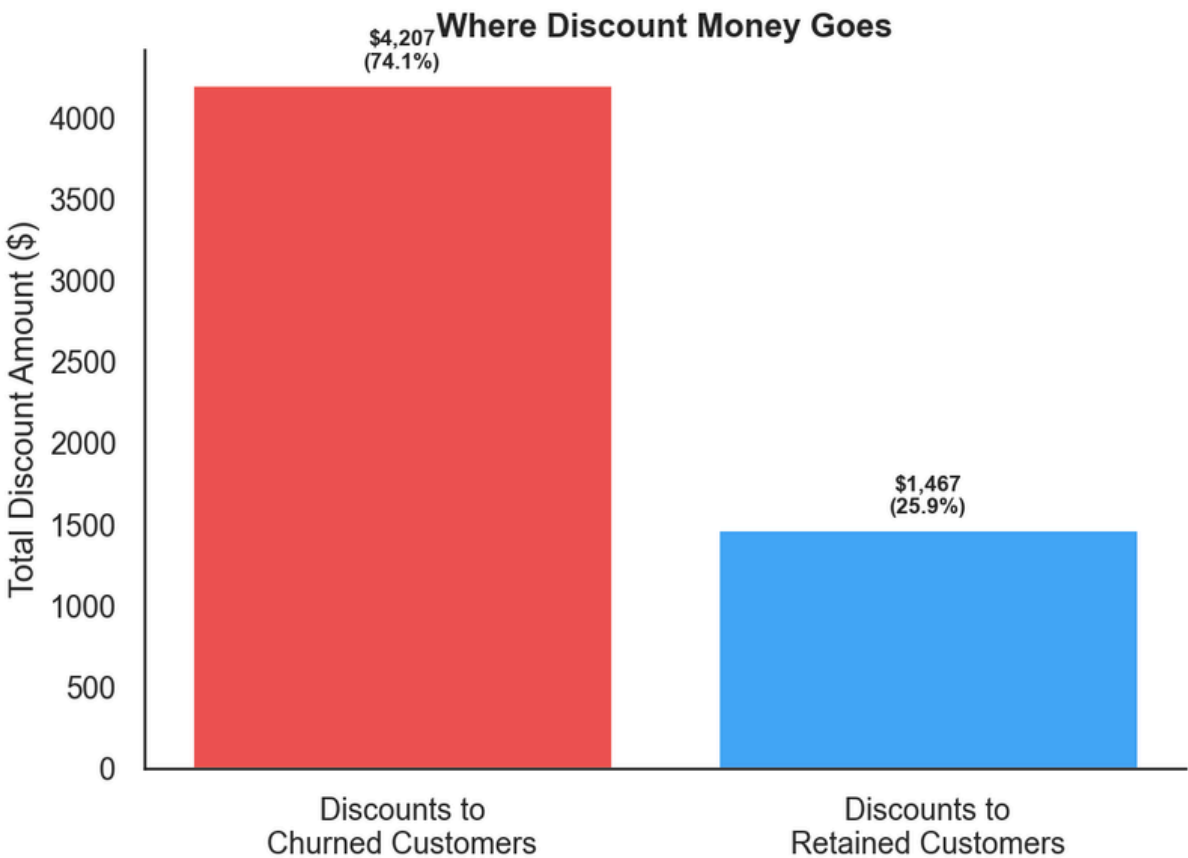
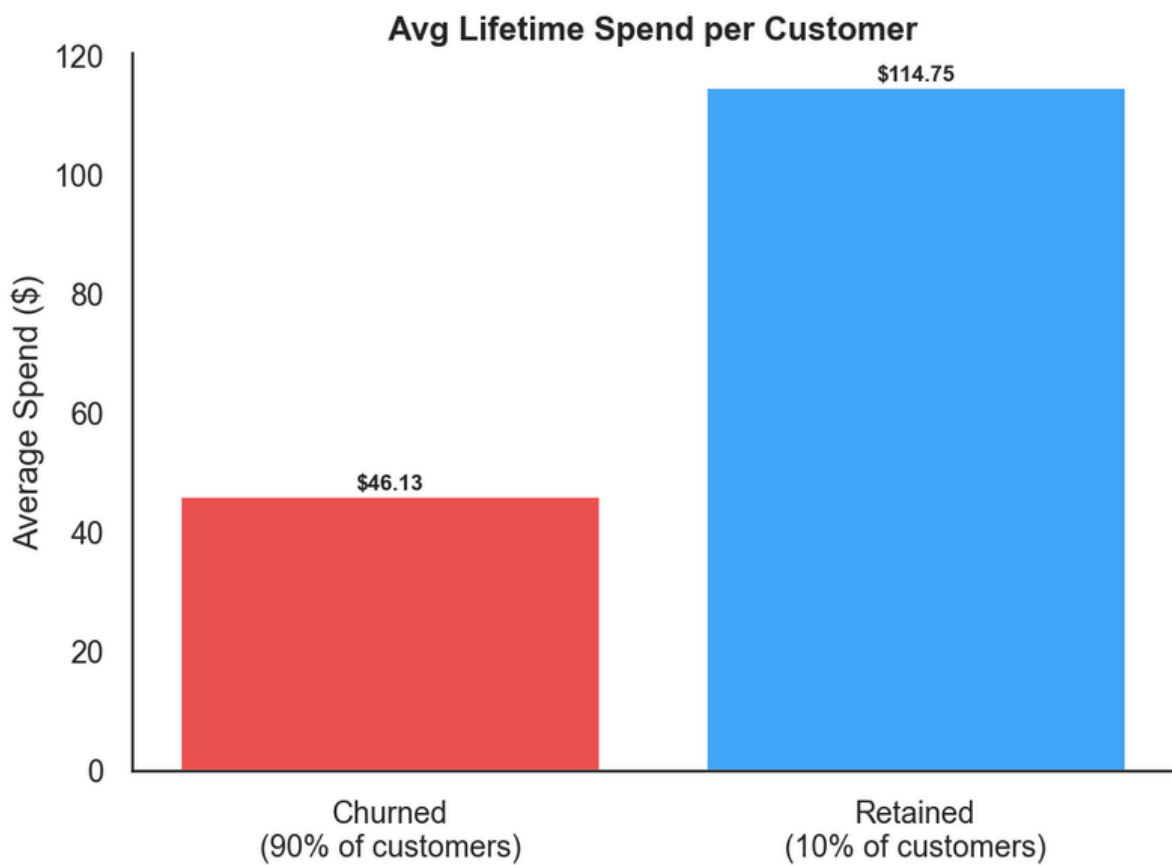
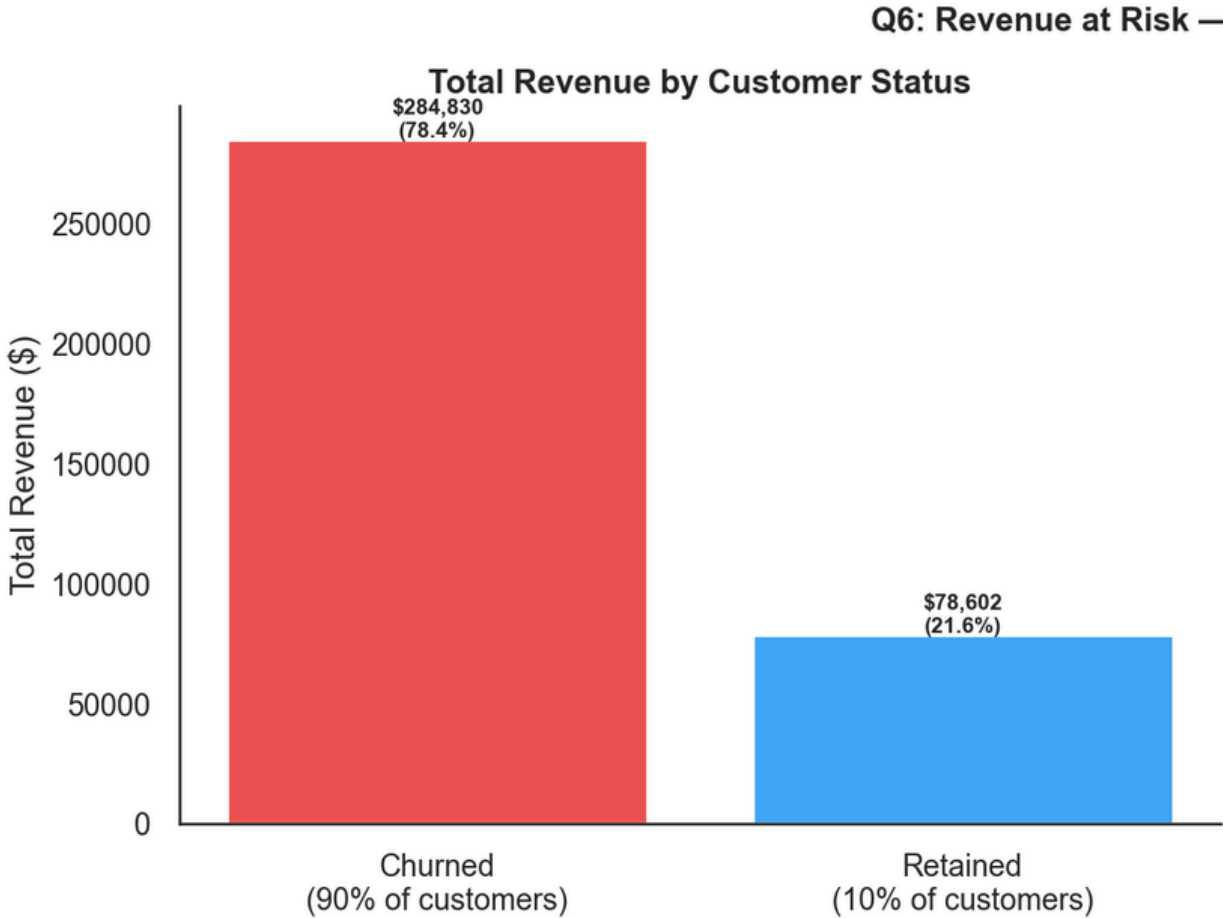


Key Insights >>>> visual deep-dive into the key metrics related to our discount strategy

Most powerful Q6 insights:

This is the most business-relevant finding of all six questions.

- 1. 90% of customers generate (78.4%) of revenue only because there are so many of them. Each individual churned customer generated only \$35.40. Each retained customer generated \$95.41.
- 2. \$4,207 of discount budget wasted. This is concrete and immediately actionable.
- 3. The recovery scenarios are conservative. A 10% retention improvement generates \$58,870, more than 10 times the wasted discount budget.
- 4. The ROI calculation is the most powerful number here. If the business redirected its wasted discount budget into effective retention tactics and converted just 10% more churned customers, it generates \$58,870. That is a 14x return on the wasted \$4,207.



Predictive Modeling

Random Forest was selected as the best model for Classification and Regression.

Classification Model - Business Question

Can we predict which customers are unlikely to return after their first purchase, based on their first order behaviour, so the business can intervene with a targeted discount at the right moment before they churn ?

Regression Model - Business Question

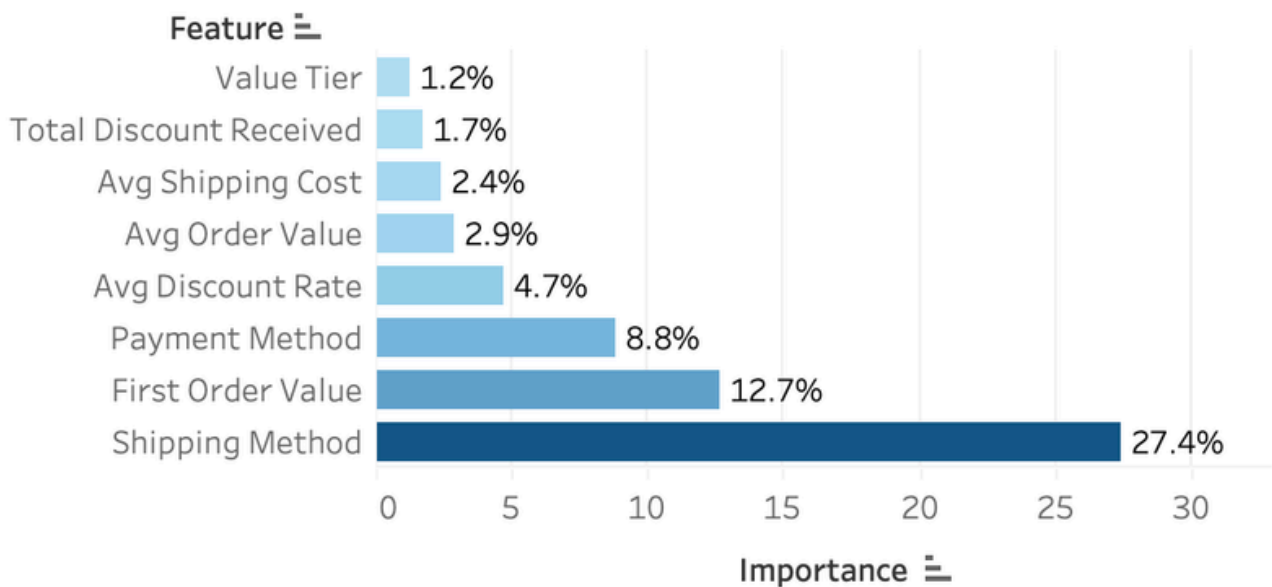
how much will a customer spend over their lifetime, and what are the key drivers of that value, so the business can identify which customers are worth giving a discount to and prioritise its most valuable customers from day one?



The Discovery

Two predictive models reveal what drives churn and lifetime value and discount behaviour is barely relevant to either.

Classification Model — What Predicts Churn?



CLASSIFICATION MODEL

AUC Score: 0.9713
Accuracy: 98.4%
Recall: 86.4%
Precision: 91.4%

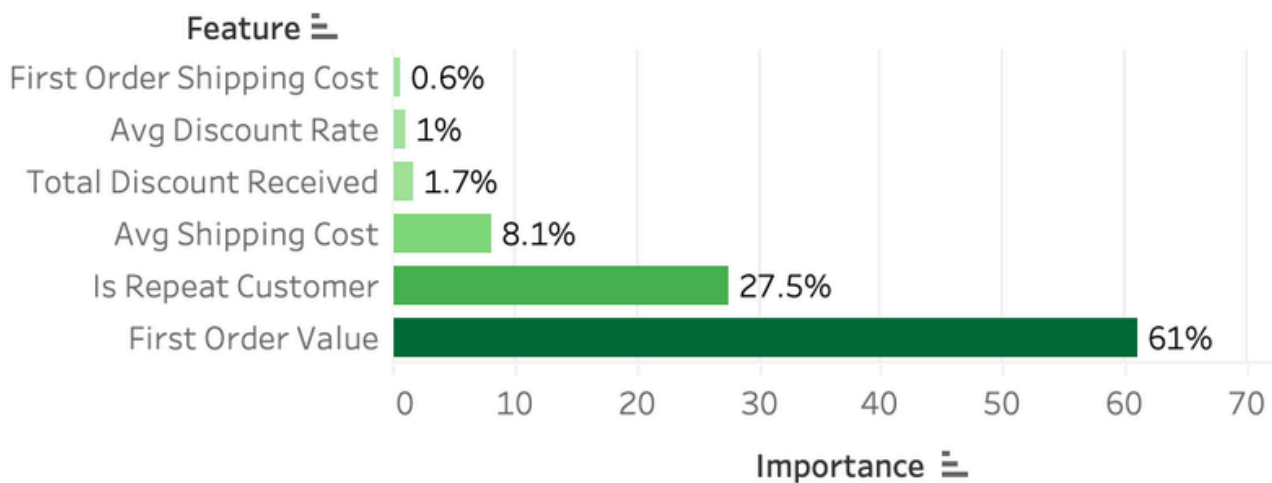
Outstanding, correctly separates churned from retained customers 97% of the time.

REGRESSION MODEL

R²: 0.6283
MAE: \$15.74
RMSE: \$31.78

Model explains 62.8% of lifetime spend variation
Average error of just \$15.74

Regression Model — What Drives Lifetime Spend?



Predicted Lifetime Value by First Order Tier

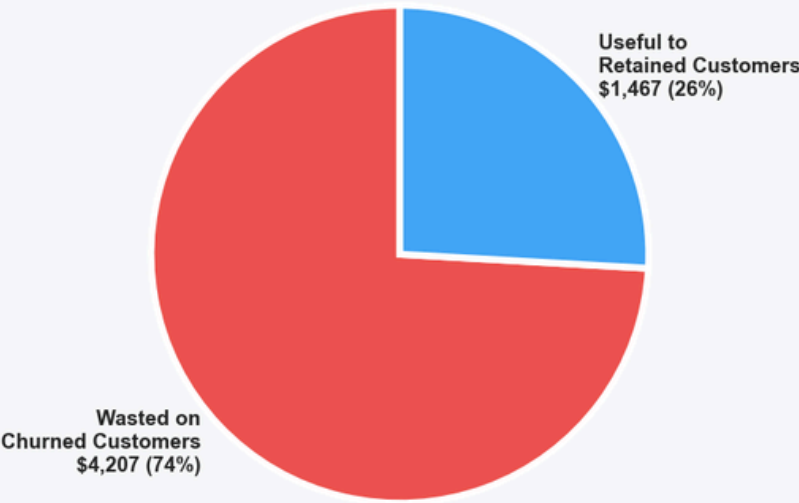


Business Impact *and* Recommendations

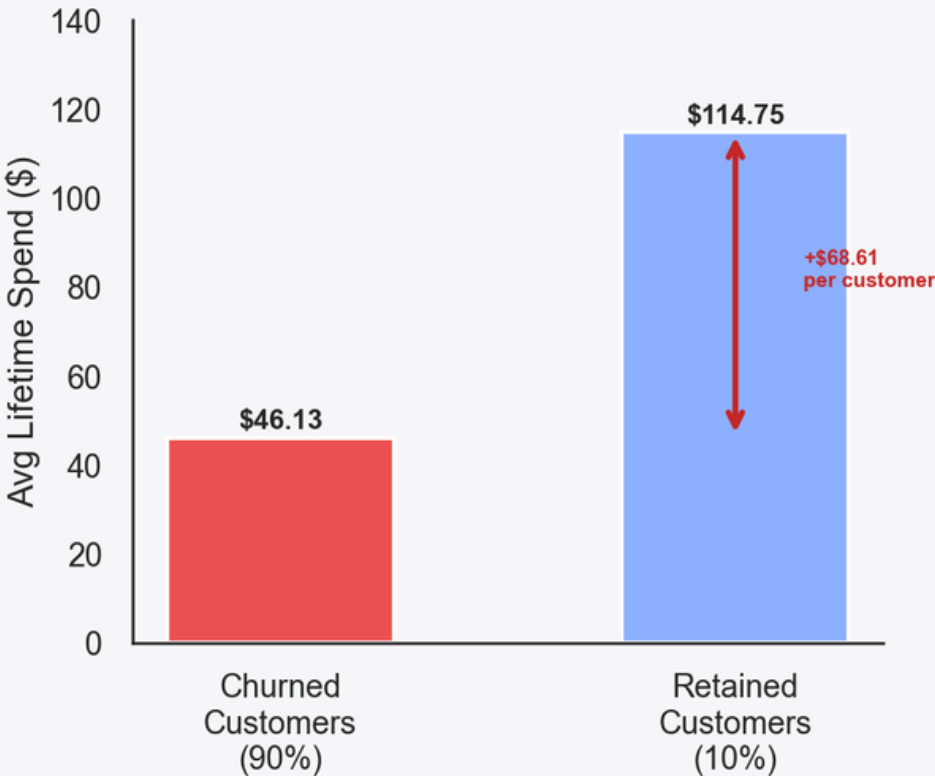
The Business Case for Smarter Discounting

How predictive modelling transforms a 5,674 discount budget into 43,497 of value

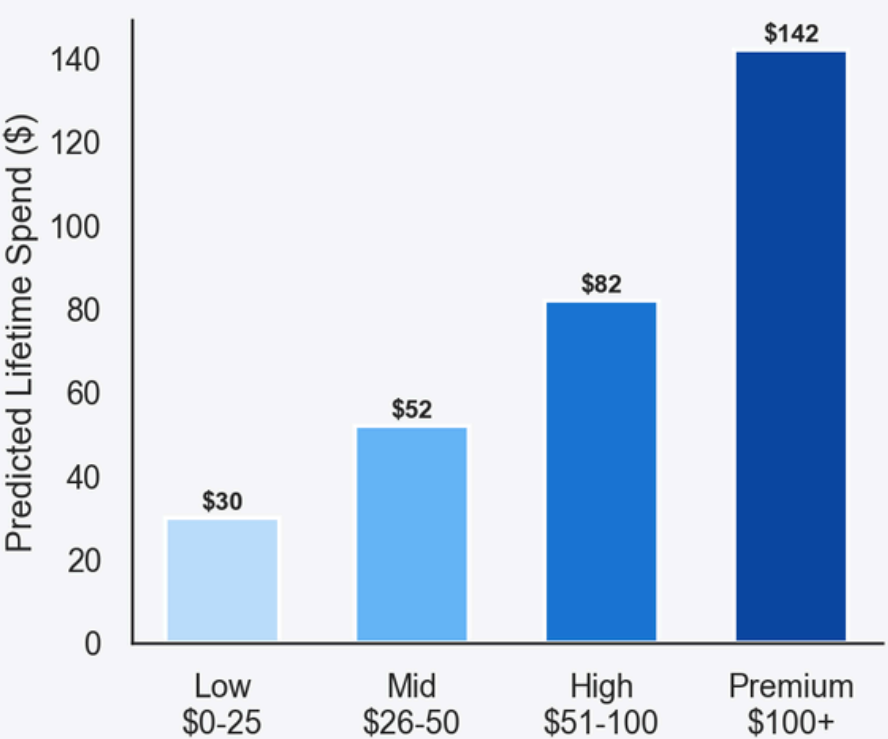
WHERE DISCOUNT MONEY GOES TODAY



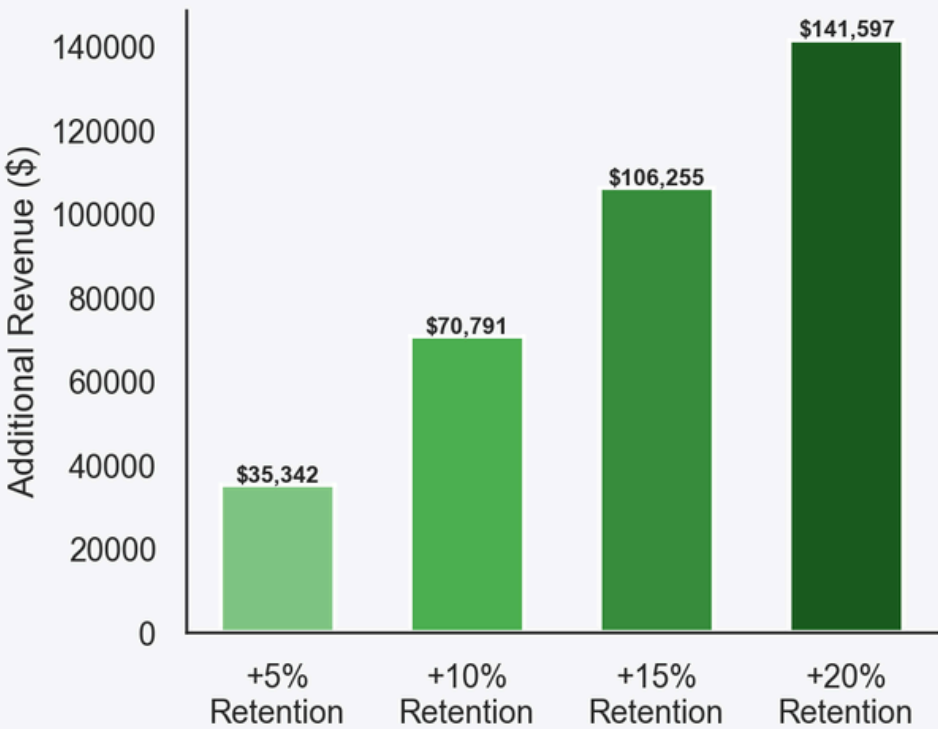
AVERAGE LIFETIME VALUE Retained vs Churned



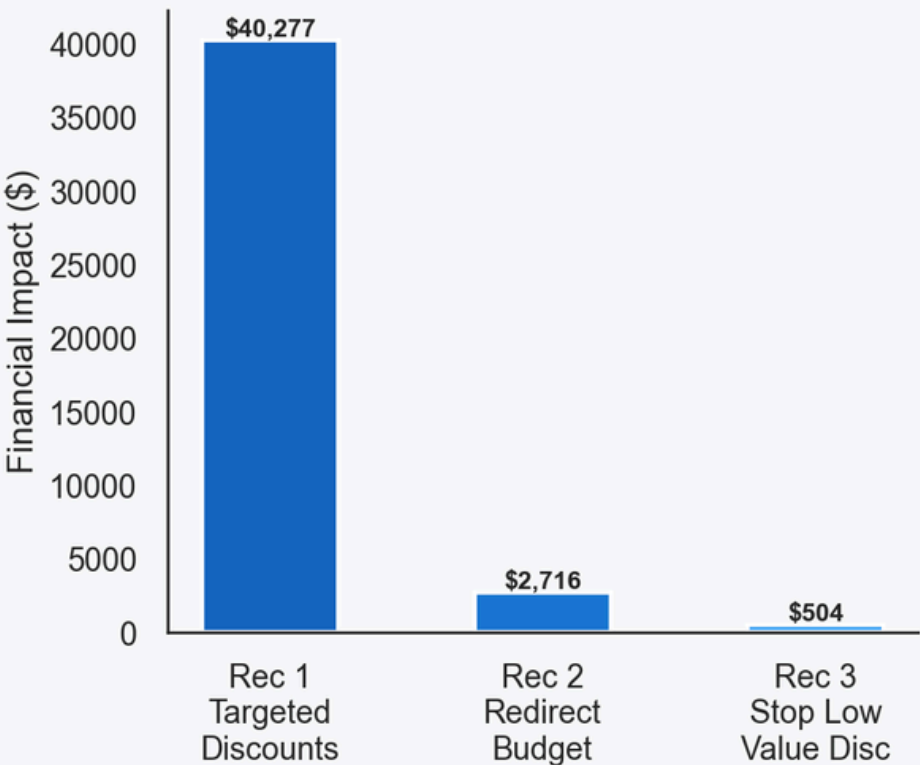
PREDICTED LIFETIME VALUE by First Order Tier



REVENUE RECOVERY POTENTIAL if Retention Improves



ESTIMATED IMPACT per Recommendation



CURRENT STATE SUMMARY

Total customers: 6,859
Churned (never returned): 6,174 (90.0%)
Retained (returned): 685 (10.0%)

Total revenue: \$363,431.52
Avg value churned: \$46.13
Avg value retained: \$114.75
Value gap per customer: \$68.61

Total discount budget: \$5,674.84
Wasted on churned: \$4,207.45 (74.1%)
Useful (retained customers): \$1,467.39 (25.9%)

BUSINESS IMPACT RECOMMENDATIONS

Recommendation 1

Target the right customers Use the model to identify high-value churned customers (first order >\$25). Converting just 10% of them generates \$40,277 in net revenue at a cost of only \$5 per customer.

Recommendation 2

Stop wasting the budget 74% of today's discount budget goes to customers who churn anyway. Redirecting just half of that waste to high-value customers generates an additional \$2,716 in net revenue.

Recommendation 3

Cut low-value discounting \$504 is currently spent discounting customers with a predicted lifetime value of only ~\$30. Stopping this immediately improves ROI with zero cost.



**Thank You
So Much for
your attention**

8 June 2026

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